© A Publication of the Centre for Research and Development (CERAD), The Federal University of Technology, Akure, Nigeria (www.journals.futa.edu.ng)

JOST 2025

Implementation of Android 14 Features on the Google Pixel 3 Through Custom ROM Development

Adewoyin, O.; Awoyemi, J.O.; Ayogu, B.A., Ayo-Ajayi, I. and Olagunju, M.

Department of Computer Science, Federal University, Oye-Ekiti, Nigeria.

Corresponding Author: Adewoyin, O.: oluwande.ajayi@fuoye.edu.ng

ABSTRACT: This study examines the development, deployment, and evaluation of a custom Android ROM for the Google Pixel 3, aiming to enhance functionality and user experience beyond the limitations of the standard Android 12 version. Built on the Android Open-Source Project (AOSP), the custom ROM integrates features from Android 14, such as dynamic lock screens, AI-driven wallpapers, and improved system updates. The research addresses the shortcomings of the stock ROM, including limited customisation options, outdated software support, and the presence of bloatware, thereby aiming to enhance performance, security, and personalisation. The development process involved establishing a cloud-based build environment, integrating device-specific code, and implementing innovative features in length. Extensive testing was conducted on both emulators and actual Pixel 3 devices to ensure functionality and stability. Performance was evaluated using benchmarking tools like AnTuTu and Geekbench, which revealed significant improvements in graphical performance; however, the custom ROM exhibited slightly lower scores in CPU-related tasks. User acceptance testing indicated positive feedback, with participants noting enhanced visual appeal, customisation options, and improved battery performance. A thorough battery analysis indicated increased power consumption in specific scenarios; nevertheless, 63.6% of users reported notable enhancements in battery life. This research contributes valuable insights into the practical challenges and advantages of custom ROM development, highlighting their potential to prolong device lifespan, offer personalised experiences, and maintain security updates beyond official support. The findings underscore the significance of user feedback in iterative design, advocating for community-driven development to create sustainable, user-centric software solutions. Future research will focus on optimising performance in CPU-intensive tasks and enhancing power management to further improve operational efficiency.

Keywords*:* Custom ROM Development, Android Open-Source Project (AOSP), Google Pixel 3, User Experience Optimisation, Mobile Device Longevity

JoST. 2025. 14(2): 43-61

Accepted for Publication, April 29, 2025

INTRODUCTION

Custom ROMs are alternative versions of the Android operating system developed to improve functionality, performance, and user experience beyond what stock ROMs offer. Their development is facilitated by Android's open-source nature, which allows modification of the Android Open-Source Project (AOSP) to suit specific needs (Pranjal et al., 2019; Shreyas, 2020). Android comprises multiple layers, including the kernel, hardware abstraction layer (HAL), native libraries, Android runtime system (ART), framework, and applications.

Several custom ROMs offer distinct features, with LineageOS recognised for stability and customisation, Pixel Experience for a clean interface and minimal bloatware, and Resurrection Remix for extensive modifications (Meetkumar, 2022). The current study

developed a custom ROM based on Pixel Experience for the Google Pixel 3, targeting Android 14. This ROM addressed specific software limitations and enhanced the device's performance. Despite newer Android versions (such as Android 15) and devices (e.g., Pixel 9) being available at the time of writing, the choice of Pixel 3 was primarily due to resource constraints during project execution. The Pixel 3 was the only test device available for development and testing, and it remains a suitable platform for demonstrating ROM customisation due to its continued support in open-source communities, reliable hardware, and compatibility with AOSP. This ROM removes manufacturer-installed bloatware, supports timely updates, and ensures broader compatibility across devices. Development involved setting up the AOSP build environment, acquiring necessary hardware binaries, syncing and modifying the source code, and conducting rigorous testing. Key features include upgrading to Android 14, integrating dynamic lock screen clocks, and introducing AI-powered wallpapers to personalise the user experience. The ROM was evaluated for functionality, stability, and performance against stock ROMs (Sutter, 2021).

Custom ROMs have been widely studied for their impact on performance, security, and user experience. Research indicates they offer better efficiency, enhanced security, and greater personalisation than stock ROMs, mainly by eliminating bloatware and introducing optimisations (Meetkumar, 2022; Suleman et al., 2020). Mudit, (2023) highlighted how AOSP enables performance enhancements, privacy-focused features, and extensive UI customisation. However, challenges such as security risks, installation difficulties, and warranty voidance remain concerns (Rahimi et al., 2019).

The Google Pixel 3 was selected for this study due to its active developer community, open-source hardware, and limited official software support. Despite being released in 2018, its hardware remains relevant, making it a strong candidate for custom ROM development. Custom ROMs provide extended software support, optimise performance, and introduce advanced customisation options, making them valuable for prolonging device usability. This study demonstrated the potential of custom ROMs in enhancing outdated devices, ensuring continued relevance and improved user experience. While it is acknowledged that Android 14 is not officially supported on the Pixel 3, this study leveraged the Pixel Experience build system and the Android Open Source Project (AOSP) to extend unofficial support to the device. Although many of the technical resources and community-driven tools are publicly available, the research contribution lies not in creating these tools from scratch, but in the structured integration, adaptation, and optimisation of these resources to solve a practical problem — extending the functional lifespan of older hardware through custom ROM development. The project demonstrates how unsupported devices can still benefit from recent Android improvements (e.g., dynamic lock screen clocks, AI wallpapers) through careful customisation, optimisation, and testing. This contributes to ongoing discourse in digital sustainability, resource optimisation, and software-level device revitalisation, especially in contexts where newer hardware is inaccessible. Furthermore, the documented development workflow, testing methodology, and performance evaluation provide a replicable model for others looking to achieve similar goals with legacy hardware.

RESEARCH METHODOLOGY

This section outlines the development process of an Android custom ROM for the Pixel 3, using AOSP as the foundation while addressing the limitations of existing custom ROMs, such as bloatware, persistent advertisements, unnecessary features, and locked bootloaders.

Development Process

The development process is comprised of the following:

Development Environment Setup: The development environment was configured using Pixel 3 resources (El-Rewini & Aafer,

2021), including the Android SDK and the AOSP version for Android 14. Due to the substantial computing resources required, the Google Cloud Platform (GCP) was utilised, providing a robust virtual machine with 64GB RAM, 8 CPU cores, and 1TB storage running on Ubuntu 22.04 LTS. Device-specific code for the Pixel 3 was obtained from the Pixel development community (El-Rewini & Aafer, 2021) and modified accordingly. The setup process was documented at https://github.com/hentaiOS-BlueCross/device_google_crosshatch.

Development and Testing: The AOSP version was acquired using the repo tool, and device-specific code was integrated. Key features implemented included various Android 14 lock screen clocks, generative AI wallpapers, and an upgrade to Android 14 for improved performance and security. Testing was conducted using emulators and real Pixel 3 devices, with all modifications and code changes systematically documented.

Hardware Constraints and Cloud Solution: A local machine with 8GB RAM and a 4-core Intel i7 processor was inadequate due to prolonged build times exceeding 24 hours (Santiago-Salazar & Rico-Bautista, 2024). Limited bandwidth further delayed synchronisation. To mitigate these constraints, a high-performance GCP virtual machine was employed, ensuring efficient compilation and development.

Tools: Various tools were used to streamline development. Repo facilitated the management of multiple Git repositories, while Git tracked code changes. GitHub and GitLab supported collaboration, and Visual Studio Code with Copilot enhanced coding efficiency. GitHub Codespaces enabled remote development, while Bash was used for build commands and file management.

Device Connectivity: The Pixel 3 was interfaced using Android Debug Bridge (ADB) for flashing builds, running tests, and retrieving logs, while Fastboot was used to flash custom ROMs and recoveries. These tools were essential for iterative testing and deployment of the custom ROM on the device.

Data Collection

Developing a custom Android ROM requires rigorous testing and feedback to enhance usability and performance on the Pixel 3. We employed benchmarking tools such as AnTuTu and Geekbench to compare system performance against the stock Android 12 version, assessing CPU and GPU speeds, memory management, and system responsiveness. For battery life analysis, Franco Kernel was used to monitor idle and active power consumption, identifying areas for optimisation. Stability and compatibility were tested through automated and manual methods to ensure seamless functionality with essential device features and popular applications.

Before public release, we conducted extensive internal testing to verify features outlined in section 2, identifying and addressing potential bugs or compatibility issues. In cases of boot loops or crashes, tools such as Android Debug Bridge (ADB), logcat, and fastboot were utilised to collect logs and diagnose errors. Troubleshooting guides and recovery methods were documented to facilitate efficient problem resolution.

Data Analysis

A quantitative approach was used to analyse benchmarking scores, battery life, and compatibility logs, guiding optimisations and feature improvements.

App Design

The design addressed limitations of existing custom ROMs, focusing on compatibility, security, and lack of official support. Key features included:

Android 14 Features – Lock screen clocks and generative AI wallpapers.

Performance Enhancements – Improved processing speed, battery life, and system efficiency.

User-Friendly Interface – Intuitive navigation for a seamless experience.

Design Decisions

AOSP Foundation – Ensured stability and compatibility.

Open-Source Development – Facilitated security updates and community collaboration.

Modular Design – Allowed flexible feature integration.

Performance Optimisation – Enhanced speed and battery life using device-specific optimisations.

User-Friendly Interface – Prioritised simplicity for ease of use.

Iterative Design and Adaptation

The design followed an iterative process, allowing refinements based on feedback and testing. Feature prioritisation was adjusted to meet user demands, security enhancements were updated to address emerging vulnerabilities, and performance tweaks were made to optimise the Pixel 3's capabilities. This approach ensured the ROM remained user-focused and adaptable throughout development.

Software Architecture

The custom ROM for the Google Pixel 3 adopted a layered architecture, a well-established approach for building complex software systems. The architecture provides modularity, maintainability, and a clear separation of concerns, as shown in Figure 1.

Figure 1 depicts the custom ROM's architecture, built on the **AOSP Core** for essential system functions. The **HAL** bridges AOSP with Pixel 3 hardware, ensuring device-specific features like camera controls and connectivity. **Device-specific code** includes drivers and kernel modifications for hardware optimisation. **Custom ROM Features** add enhancements such as advanced camera controls and privacy settings. The **User Interface (UI)** builds on AOSP, offering a customised user experience.

Component Interactions and Responsibilities

The **AOSP core** communicates with the **HAL**, translating requests into device-specific code commands via drivers and kernel modifications. **Custom ROM features** leverage AOSP and may interact with HAL or device code for hardware functions. The **UI layer** connects with underlying layers to display information, process input, and offer customisation.

Programming Languages

By focusing on the limitations and desired features, the primary programming languages used for custom ROM development are:

Java: The core functionalities of the ROM leveraged Java, as it is the primary language used for Android development and interacts heavily with the AOSP core.

C/C++ (Optional): For deeper system-level interactions or performance optimizations, particularly when integrating device-specific code (kernel modifications, drivers), C/C++ was used.

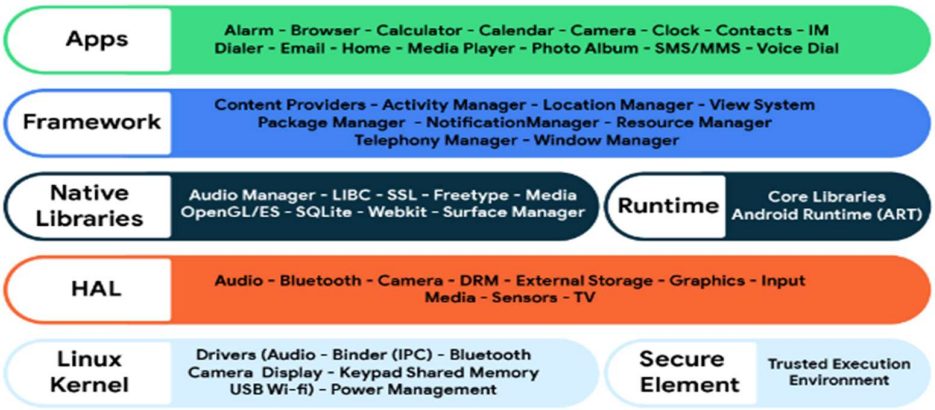


Figure 1: Software Architecture for the custom ROM. Source: Google. (2023, July 10). Android documentation. Retrieved from https://source.android.com/docs/setup/about

Tools and Frameworks

Android SDK: Provides libraries, tools, and emulators for ROM development, including AOSP downloads, building, and testing.

Build Tools (repo, etc.): Manages and builds the ROM source code, with **repo** handling AOSP codebase management.

Version Control (Git): Tracks changes, enables rollbacks, and supports collaboration.

Debugging & Profiling: Android Studio's debugger and profiling tools help identify bugs and optimise performance.

Build Process

The build process for the custom ROM are:

Cloud-Based Environment: A Ubuntu VM on GCP (64GB RAM, 8 CPUs, 1024GB storage) handled the build efficiently.

AOSP Device Tree Setup: The Pixel 3 device tree was cloned, updated with LineageOS changes for Android 14 compatibility, and conflicts resolved.

ROM Customisation: Modifications were made to integrate features, and build scripts were adjusted for Android 14.

Build & Debugging: Initial builds were tested; bootloop issues were diagnosed using ADB logcat and fastboot, followed by iterative fixes.

Bug Fixing & Feature Implementation: Testing on emulators and real devices ensured stability while integrating Android 14 features like lock screen clocks and AI wallpapers (Figures 2–4).

Lock screen designs

The lock screens are prebuilt system apps provided by Google but not integrated into Android. The lock screens vary from each other in terms of the design features incorporated. The design features are fonts, colour and position. The lockscreens are shown in Figure 2a-i.

Figure 2a-i shows the nine lock screen designs for the developed custom ROM. The design features for the lock screens are explained in terms of visuals and layouts, background, time display, customisation options, and suitability for Android lock screen.

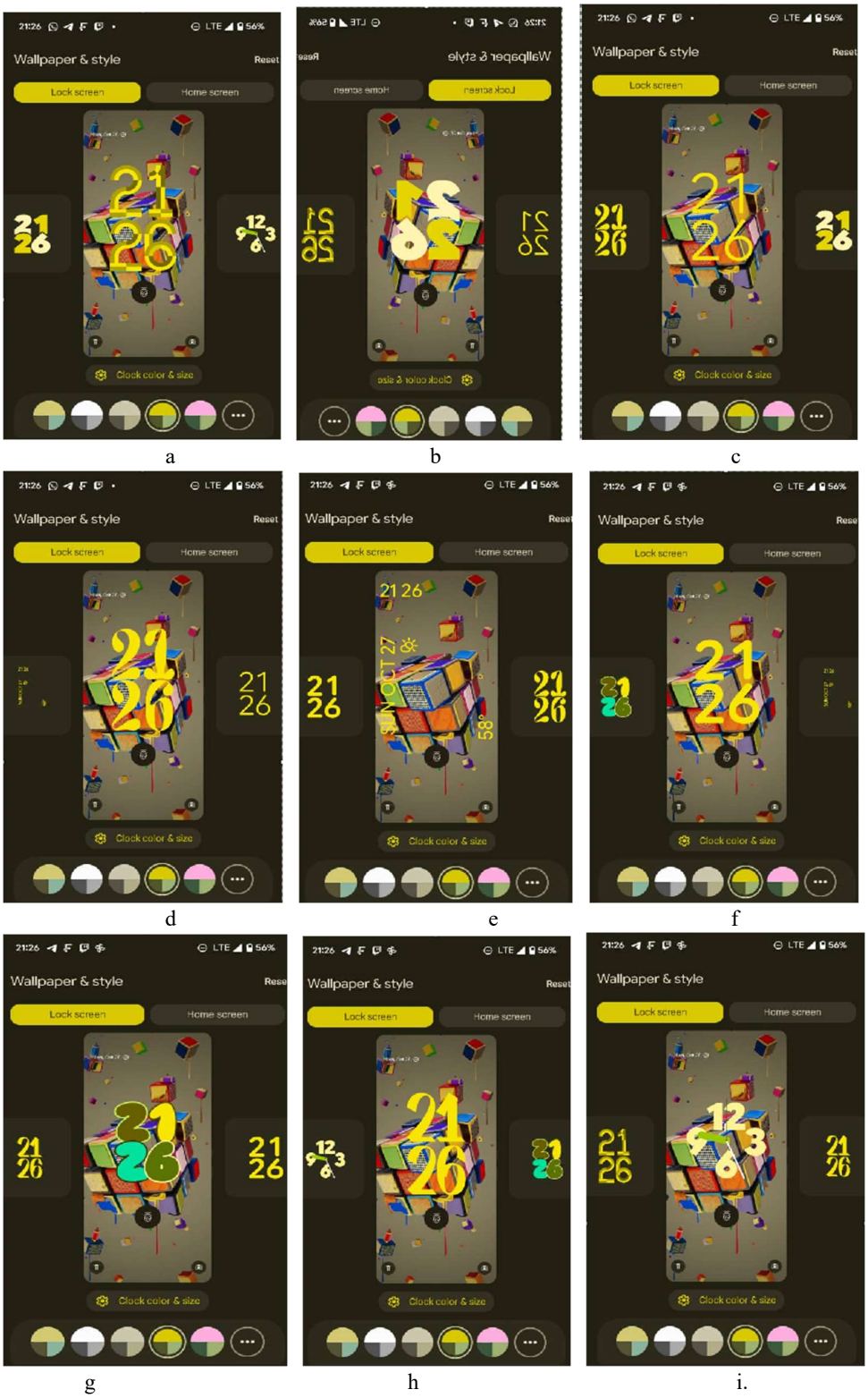
Design A features a vibrant, abstract background with geometric shapes, creating a dynamic look. The time is displayed in a bold yellow font with large overlapping numbers for easy readability. Users can customise the clock's colour and size to suit their preferences. This design is ideal for those who appreciate modern, visually striking aesthetics while ensuring quick time visibility.

Design B features a colourful, abstract background with geometric shapes, similar to Design A, but with a bold white font for the time display. The large overlapping numbers maintain a modern look, with slight font variations. Users can customise the clock's colour and size, making it ideal for those who prefer a minimalist yet visually engaging interface that keeps the focus on time readability.

Design C features a vibrant background with colourful 3D geometric shapes on a beige backdrop, creating a modern and dynamic look. The time, "21:26," is displayed in large, yellow, stylised numbers for easy readability. Users can customise the clock's colour and size to personalise their lock screen. Design C is ideal for users who prefer a colourful, dynamic lock screen with a modern touch. The bold, stylised numbers ensure quick and clear readability.

Design D maintains visual consistency with Design C, featuring a vibrant beige backdrop with colourful 3D geometric shapes. The time, "21:26," is displayed in large, yellow, stylised numbers at the centre, with slight layout variations. Users can customise the clock's colour and size for personalisation. Ideal for those who value consistency, it balances aesthetics with clear readability.

Design E features a vibrant, abstract background with geometric shapes like cubes and rectangles, creating a dynamic and engaging appearance. The time is displayed in yellow at the top left corner, with the date and temperature arranged vertically on either side for a sleek, modern touch. The inclusion of weather conditions, indicated by a sun icon and temperature, enhances functionality, allowing users to check essential information at a glance. Users can customise the clock's colour and size to personalise their lock screen. This design is ideal for those who value quick access to time, date, and weather while appreciating a visually appealing, modern aesthetic.



a b c

d e f

g h i.

Figure 2a-i: Lock screen designs

Design F features a vibrant, abstract background with geometric shapes, maintaining a consistent visual theme with Design E. The time is displayed in large yellow numbers at the centre, making it the focal point for instant recognition. With a minimalist approach, it omits additional details like the date and temperature, creating a clean and uncluttered design. Users can customise the clock's colour and size to personalise their lock screen. This design is ideal for those who prefer a straightforward interface with a strong visual appeal, combining simplicity with a colourful, engaging background.

Design G features a playful and colourful wallpaper with various geometric shapes, creating a vibrant and dynamic look. The time is displayed in large, overlapping numbers—yellow for "21" and green for "26"—for a bold, eye-catching effect. Users can customise the clock's colour and size to match their preferences. Ideal for those who appreciate a lively lock screen, this design balances visual appeal with readability, though the overlapping numbers may slightly hinder quick time recognition. The customisation options enhance user engagement by allowing personalisation to suit individual tastes.

Design H features a playful and colourful wallpaper with geometric shapes, similar to Design G, but with a clearer time display. The large, yellow numbers are presented in a traditional, non-overlapping style for easy readability. Users can customise the clock's colour and size, allowing for some personalisation while maintaining a clean and straightforward interface. This design is ideal for those who prioritise clarity, as the bold numbers ensure quick time recognition. While it retains a conventional look, the colour customisation adds a touch of flexibility to suit individual preferences.

Design I features a sleek black background with subtle geometric patterns, creating a minimalist and sophisticated look. The time is displayed in large, bold yellow numbers at the centre, ensuring excellent readability. Users can customise the clock's colour and size, allowing for personalisation while maintaining a refined aesthetic. Ideal for those who appreciate elegance, this design combines simplicity with a touch of class. The clear, bold numbers enhance functionality, making it easy to check the time at a glance, while customisation options provide flexibility to match individual preferences.

Wallpaper designs

The AI Wallpaper is a prebuilt system app by Google but not integrated into Android. Like the lock screens, the wallpapers differ in font, colour, and position. Figure 3a-h displays these variations.

Design A features a surreal bicycle composed of pink and purple flowers, creating a whimsical and dreamlike aesthetic. This artistic imagery adds a sense of fantasy and elegance to the device. The dominant pink and purple hues contribute to a calming and visually appealing effect, while the floral elements enhance its beauty. It is ideal for creative users, this wallpaper offers a unique and personalised touch, appealing to those who appreciate imaginative visuals. The soothing colours and floral motifs evoke positive emotions, making it a great choice for users seeking a serene and pleasant visual experience.

Design B is a volcanic Wallpaper showcasing a striking image of white lava with vibrant hues, resembling a volcanic landscape. The gradient of colours adds depth and intensity, creating a visually dynamic effect. The contrast between the white lava and the vivid background enhances its impact, making it an eye-catching design. Ideal for adventurous users, this wallpaper exudes energy and excitement, appealing to those who enjoy bold and dramatic visuals. The high-contrast colours and dynamic composition ensure a strong visual presence, making it a perfect choice for users seeking a memorable and engaging background.

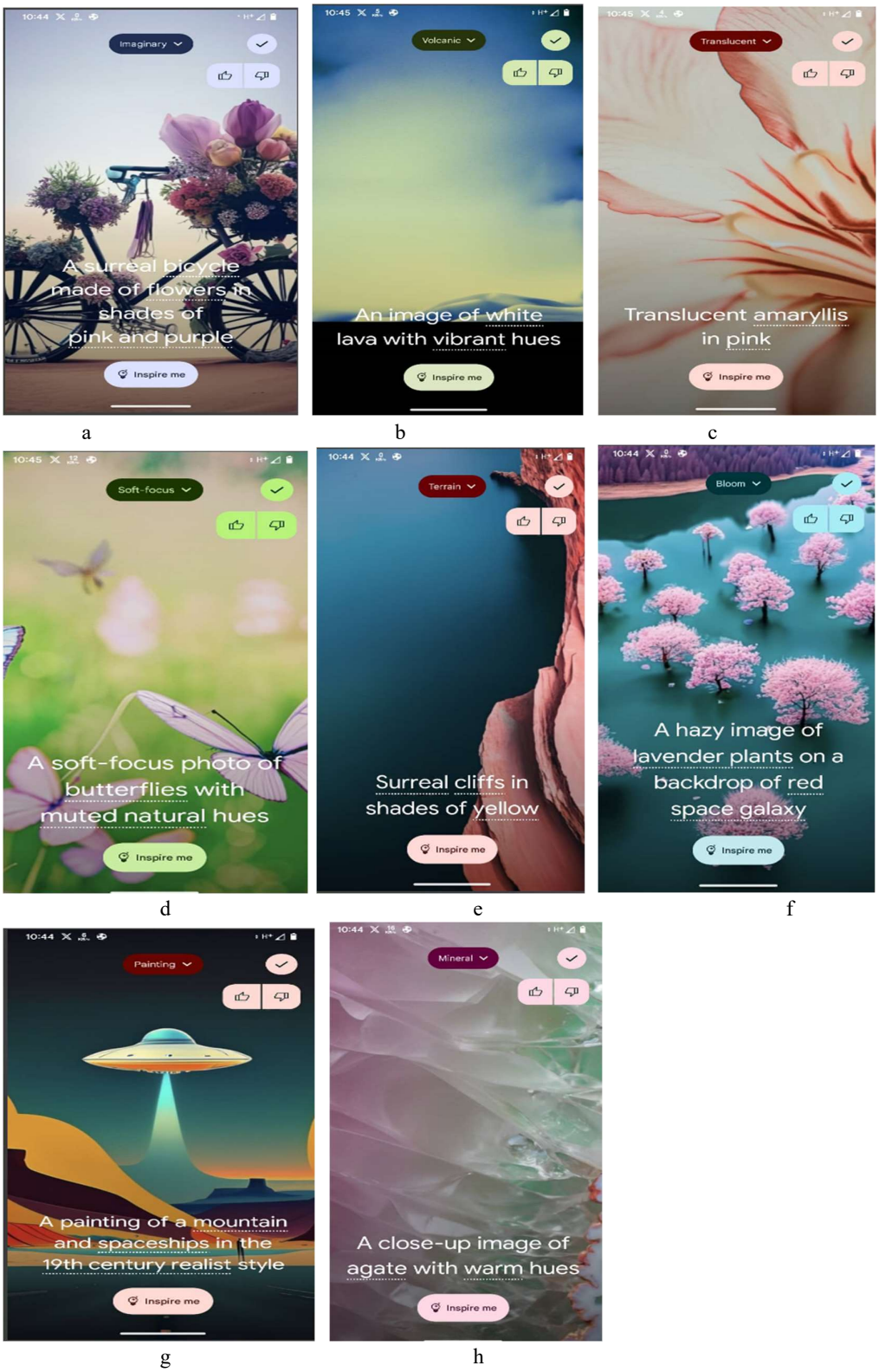


Figure 3a-h: Wallpaper designs

Design C is a translucent Amaryllis Flower, features a close-up image of a delicate pink amaryllis flower, highlighting its intricate petals and natural elegance. The soft pink hues create a calming and soothing effect, making it an ideal choice for users who appreciate subtle aesthetics and floral beauty. Perfect for nature lovers, this wallpaper adds sophistication and tranquillity to the device. Its serene colour palette enhances relaxation, making it a visually pleasing option for those who prefer a gentle and elegant background.

Design D features a soft-focus photograph of butterflies with muted natural hues, creating a dreamy and ethereal effect. The blurred background enhances the whimsical feel, while the butterflies remain in focus, adding a delicate touch of nature. The harmonious colour palette ensures visual appeal without being overly distracting. Ideal for dreamers and nature enthusiasts, this wallpaper exudes subtle elegance, making it perfect for users who appreciate refined and enchanting imagery with a gentle, understated aesthetic.

Design E features a surreal landscape with vibrant yellow cliffs set against a deep blue body of water, creating a striking contrast that enhances depth and tranquillity. The bold interplay of warm and cool tones makes the wallpaper visually captivating and aesthetically appealing. Ideal for nature enthusiasts, this design celebrates natural landscapes while adding an artistic touch to the device. The serene water combined with the dynamic cliffs evokes a sense of calm and inspiration, making it a perfect choice for users seeking both visual impact and soothing ambience.

Design F features a hazy image of lavender plants set against a red space galaxy, blending natural and cosmic elements to create a dreamy and otherworldly aesthetic. The contrast between the soft lavender tones and the vibrant red galaxy background enhances its ethereal and captivating visual appeal. Ideal for dreamers and stargazers, this wallpaper combines the beauty of nature with the mystery of space, making it both visually striking and thought-provoking. Its unique composition serves as a source of inspiration and creativity, making it a standout choice for users who appreciate imaginative and engaging backgrounds.

Design G showcases a captivating 19th-century realist-style painting of a mountain and spaceships, blending historical art with modern science-fiction elements. The artwork features a UFO emitting a beam of light against a stylised mountain and gradient sky, creating a striking and imaginative composition. With natural tones for the mountains and sky contrasting against the surreal spaceships, this wallpaper appeals to art enthusiasts who appreciate unique and conversation-starting visuals. Its detailed and realistic painting technique adds sophistication and creativity to the device, while the "Inspire me" button suggests it is designed to evoke curiosity and wonder, making it ideal for users seeking a visually stimulating background.

Design H: Mineral features a close-up image of agate with warm hues, showcasing intricate patterns and colours that highlight the natural beauty of the mineral. The colour scheme, ranging from deep reds to soft oranges, creates a harmonious and soothing palette that is easy on the eyes. Ideal for nature lovers, this wallpaper appeals to users who appreciate the elegance of natural elements, offering a serene and relaxing visual experience. The warm, earthy tones enhance its calming effect, making it perfect for those seeking a tranquil and refined background that seamlessly blends aesthetic elegance with nature-inspired design.

Figure 3a-h shows the nine lock screen designs for the developed custom ROM. The design features for the AI wallpapers are explained in terms of visuals and aesthetics, and suitability for Android AI wallpaper in subsequent sections.

Android 14 Upgrade

After integrating the new lock screens and wallpapers, at the development time, android 14 was the recent android version available on the AOSP platform. We made several changes to the device-specific code and built and boot Android 14 on the device, making it an upgrade to Android 14. The upgrade is shown in Figure 4.

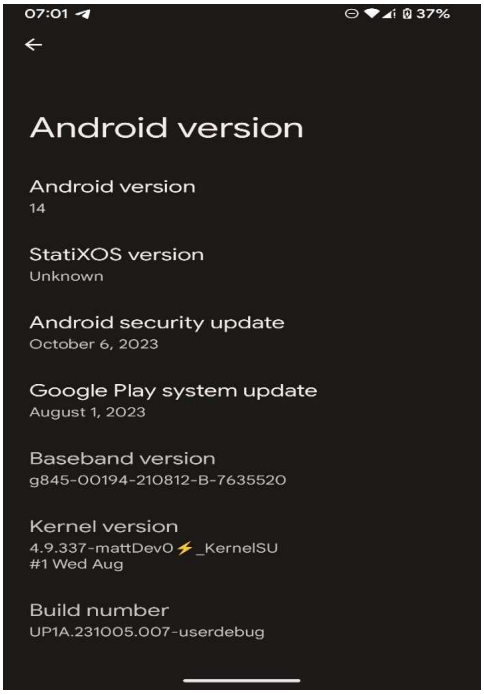


Figure 4: Android 14 upgrade screen

Figure 4 displays the successful installation of a custom ROM upgrading the device to Android 14. The screen confirms the update, incorporating Google's latest features and improvements. The custom ROM includes security updates as of October 6, 2023, ensuring protection against recent vulnerabilities. The kernel version, developed by "mattDev0" with KernelSU, suggests performance enhancements and potential root access for advanced customisation. Additionally, the build number indicates a userdebug build, typically used for development and debugging, implying the presence of extra developer tools and capabilities.

With Android 14, users can expect improved functionality, optimisations, and possibly unique features introduced by the custom ROM. Custom ROMs like StatiXOS often provide greater personalisation options than stock ROMs, allowing users to tailor their devices to their preferences. This enhances user experience by offering flexibility in system settings, interface customisation, and performance tuning.

Testing

Testing is a crucial aspect of software development, and this study employed unit and system testing. Unit testing focuses on verifying individual software components in isolation, such as functions, methods, or procedures (Taley & Pathak, 2020). It ensures that each unit performs as expected and is typically automated for efficiency. A major benefit is early defect detection, reducing bug resolution costs and supporting code refactoring (Khorikov, 2020). In this study, unit testing was conducted on individual components of the custom ROM, including device drivers, kernel modules, and feature implementations, using automated tests and test harnesses.

System testing evaluates the entire software system as a cohesive unit, ensuring it meets specified requirements and functions correctly in an integrated environment (Chahar & Bhatia, 2025). It identifies issues related to compatibility, security, and performance ((Taley & Pathak, 2020). The custom ROM underwent comprehensive system testing on emulators and real Pixel 3 devices. Following Savola et al., (2020), the ROM's effectiveness was assessed using performance, functionality, stability, and user experience metrics. These provided both qualitative and quantitative insights, enabling comparisons with stock Android 12 and other custom ROMs.

Performance Metrics

Benchmark scores were used to assess the custom ROM's performance, following Bissig et al., (2018). **AnTuTu** evaluates overall hardware performance, including CPU, GPU, memory, and storage. Its CPU benchmark tests mathematical calculations, algorithms, and multi-core processing, while the GPU benchmark assesses graphical performance through five tests. Additionally, AnTuTu measures AI-related tasks like image classification (Antutu, 2024). **Geekbench**, developed by Primate Labs, measures CPU and GPU performance across platforms. It includes single-core and multi-core tests, with Geekbench 6 incorporating productivity, developer tasks, machine learning, and image synthesis (Geekbench, 2024).

Experimental Setup

To ensure comprehensive evaluation, the ROM was tested in both emulated and real-world environments. It was flashed onto Pixel 3 devices to assess real-world performance, including network connectivity, camera usage, and general usability. Benchmarking applications (AnTuTu, Geekbench 6) provided performance data. **User tasks** included measuring boot time, app launch time, and battery life across various activities such as web browsing, video playback, gaming, and calls. Stress testing involved running multiple applications and graphically demanding tasks to evaluate stability under heavy load.

User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted with 13 Pixel 3 users of varying technical expertise to assess usability, satisfaction, and potential issues. Surveys with researcher-defined questions evaluated user experience, functionality, and performance specific to custom ROMs. The findings provided insights into the ROM's appropriateness and areas for enhancement. For future deployment, more detailed user testing approaches are recommended to refine performance and stability further. This comprehensive evaluation ensures a well-rounded assessment of the ROM's effectiveness and highlights areas for improvement.

RESULTS AND DISCUSSION

This section presents the evaluation of the custom Android ROM for the Google Pixel 3, comparing performance metrics from benchmarking and user acceptance testing against the stock ROM and alternatives.

Performance Results

Performance was assessed using AnTuTu and Geekbench, as shown in Table 1. The transition to Android 14 introduces additional features and visual enhancements, increasing resource demands compared to the stock Android 12.

Stock Android 12 leads in overall system performance with the highest AnTuTu score (450,000), benefiting from optimised resource management. LineageOS Android 14 follows at 435,000, while Pixel Experience scores 425,000. The custom ROM scores the lowest at 420,000, suggesting potential optimisation gaps.

In Geekbench 6 single-core performance, Stock Android 12 again ranks highest (561), followed by LineageOS (490), Pixel Experience (447), and Custom ROM (434), indicating weaker efficiency in single-threaded tasks. Multi-core performance follows a similar pattern, with Stock Android 12 leading (1908), ahead of LineageOS (1639), Pixel Experience (1575), and Custom ROM (1409), reflecting its lower multi-core efficiency.

Table 1: Performance Benchmark Comparison of Custom and Stock ROMs on Google Pixel 3

Benchmark	Custom ROM Android 14	Stock Android 12	LineageOS Android 14	Pixel Experience Android 14
AnTuTu (Overall)	420,000	450,000	435,000	425,000
Geekbench 6 (Single-Core)	434	561	490	447
Geekbench 6 (Multi-Core)	1409	1908	1639	1575
Geekbench 6 (GPU)	3500	1651	3400	3300

However, the Geekbench 6 GPU benchmark presents a different trend. The custom ROM outperforms all others (3500), suggesting a focus on graphical optimisation, making it suitable for gaming and visually intensive tasks. LineageOS (3400) and Pixel Experience (3300) also show strong GPU performance, while Stock Android 12 ranks lowest (1651), indicating less emphasis on graphical processing.

Overall, Stock Android 12 provides the most balanced performance, excelling in CPU tasks. The custom ROM, while trailing in CPU benchmarks, stands out in GPU performance,

making it ideal for graphics-heavy applications. LineageOS offers a well-rounded option, whereas Pixel Experience, though competitive, does not dominate in any category. These findings highlight the trade-offs between ROMs based on performance priorities.

Boot and App Launch Times

The boot times for the custom ROM was compared with stock 12. The results in Table 2 shows the boot and app launch durations for the custom ROM and Stock Android 12. The comparison between the Custom ROM Android 14 and Stock Android 12 reveals notable differences in both boot time and average app launch time, suggesting varying levels of optimization for each ROM.

The boot time measurement shows that Stock Android 12 outperforms the custom ROM, with a significantly faster boot time of 40 seconds compared to 60 seconds for the Custom ROM. This 20-second difference indicates that Stock Android may have better system-level optimizations, possibly through refined boot processes, fewer start-up services, or more efficient handling of background tasks during the start-up phase. The longer boot time for the Custom ROM suggests that it may require further optimisation to reduce system load during boot, which could involve disabling unnecessary start-up services or streamlining initial processes.

When considering the app launch time, Stock Android 12 again demonstrates superior performance, with an average launch time of 0.4 seconds, which is more than twice as fast as the 0.9 seconds recorded for Custom ROM Android 14. This disparity suggests that Stock Android's optimisations extend beyond the boot phase, providing faster access to applications. It may be indicative of better memory management, efficient use of system resources, or enhanced preloading of frequently accessed applications. The slower app launch times observed in the Custom ROM may imply heavier system overhead or less efficient resource allocation, potentially affecting user experience in everyday use.

The overall findings suggest that Stock Android 12 is better optimised for both system start-up and application performance, which may appeal to users seeking a more responsive and efficient system. The Custom ROM, while potentially offering other features or customisations, may require additional optimisation efforts to improve its competitiveness in these key performance metrics. Such efforts could include refining system services, better memory management strategies, or optimising app loading mechanisms to reduce delay. This analysis highlights the importance of performance tuning in custom ROM development to enhance the end-user experience.

Battery life

The battery life results for various usage scenarios between Custom ROM Android 14 and Stock Android 12 are summarised in Table 3.

Table 2: Benchmark Performance Metrics of Custom and Stock ROMs on Google Pixel 3

Metric	Custom ROM Android 14	Stock Android 12
Boot Time	60 seconds	40 seconds
App Launch Time (Average)	0.9 seconds	0.4 seconds

Table 3: Battery Life Comparison of Custom ROM Android 14 and Stock Android 12 on Google Pixel 3 Across Usage Scenarios

Usage Scenario	Custom ROM Android 14 (Hours)	Stock Android 12 (Hours)
Idle	37	48
Web Browsing	4	6
Video Playback	3	5
Gaming	2	3.5
Call Time	4	6

The Table 3 compares the battery life of Custom ROM Android 14 and Stock Android 12, showing that Stock Android 12 consistently outperforms the Custom ROM in various scenarios. These differences suggest variations in power management and system resource optimisation between the two versions.

In idle mode, Stock Android 12 lasts 48 hours, 11 hours longer than the Custom ROM, likely due to better background process management. During web browsing, Stock Android achieves 6 hours compared to 4 hours for the Custom ROM, indicating more efficient resource utilisation. For video playback, Stock Android extends battery life to 5 hours, while the Custom ROM lasts only 3 hours, suggesting better media playback optimisation. Gaming highlights a 1.5-hour difference, with Stock Android reaching 3.5 hours versus 2 hours for the Custom ROM, possibly due to superior GPU and power management. In call time tests, Stock Android provides 6 hours, 2 hours more than the Custom ROM, reflecting better network and voice communication efficiency.

Overall, Stock Android 12 offers longer battery life across all scenarios, making it a better choice for users prioritising efficiency. The Custom ROM, while potentially offering customisation benefits, consumes more power and could improve through better background service handling, power-saving settings, and resource management. This analysis highlights the need for effective power management strategies in ROM development to enhance device usability.

User acceptance testing

User acceptance testing involves testing a software product with end users in order to obtain acceptability and another perspective on a developed software (Dharta et al., 2024).

A select group of Pixel 3 users with experience using custom ROMs will be invited to participate in user acceptance testing (UAT). They installed and used the custom ROM in their daily lives, providing valuable feedback on its functionality, usability, and overall experience. This feedback was used to further refine and improve the ROM before its wider release. The results are described in subsequent sections.

Experience Duration: The number of participants based on how long they have been using custom ROMs is categorized in the pie chart in Figure 5.

The chart shows that 33.3% of users have been using custom ROMs for 1-2 years, 8.3% have been using them for less than a year and 3-5 years, while 50% of the users have been using custom ROMS for more than 5 years. The pie chart highlights that the majority of users are relatively experienced, with a significant proportion having 1-2 years or more than 5 years of usage.

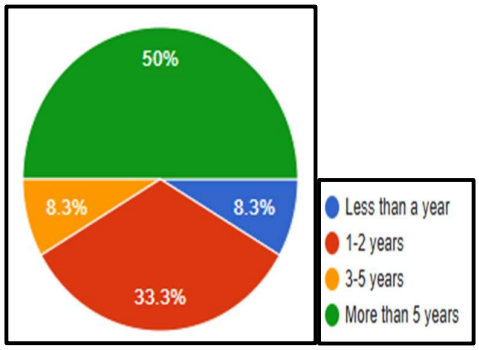


Figure 5: Duration of Use by Pixel 3 Participants

Primary reasons for using a custom ROM

The primary reasons given by users for adopting a custom ROM were categorized into performance, the latest android version, customisation, privacy, Pixel-only features, and others. The result in Figure 6 shows that 75% of the users adopt custom ROM because of performance. 83% adopted because of the latest Android version, 25% adopted because of customization, 8.3% adopted because of Pixel-only features, 8.3% adopted because of other related features and nobody adopted because of privacy.

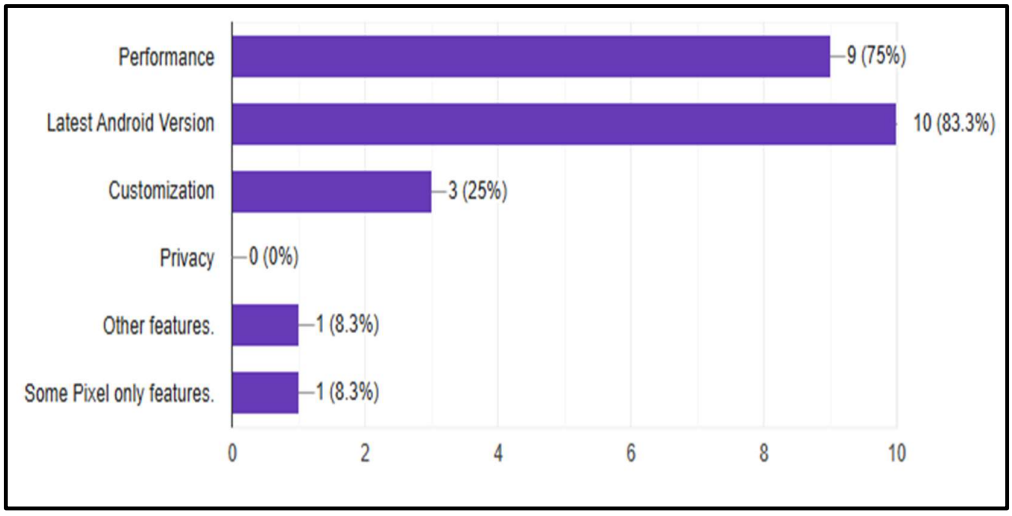


Figure 6: Primary reasons for using a custom ROM

Performance Rating Compared to Stock ROM

Users rated the overall performance of the custom ROM compared to the stock ROM on a scale of 1-5, with 5 being excellent and 1 being poor. Figure 7 shows the performance of the custom ROM as 50% of the participants indicated a rating of 4 while 50% indicated a rating of 6, signifying a positive performance of the custom ROM.

Improvements in Battery Life

The survey inquired whether users noticed any significant battery life improvements compared to the stock ROM. Figure 8 shows that 63.6% indicated a 'Yes', 18.2% indicated a 'No', 9.1% indicated 'minor improvements and 9.1% indicated 'no improvement' for battery lives. Overall, the battery lives of devices are being preserved with the new custom ROM.

Performance Issues Encountered

Participants were asked if they had encountered any performance issues, such as slowdowns, crashes, or app compatibility problems. The bar chart in Figure 9 visualizes the distribution of performance issues encountered by custom ROM users. Five users reported experiencing performance issues such as random reboots, slowdowns, crashes, battery problems, and RAM management limitations, while the other five users did not report any performance-related issues. The data indicates a split where half of the users encountered problems, highlighting the variability in user experiences with custom ROMs

Features Considered Most Valuable

Users were asked to select features that they found most valuable in the custom ROM. The bar chart in Figure 10 illustrates the features considered most valuable by users in the custom ROM. "Android 14 Lock Screen Clock" and "Other Features (e.g., Face Unlock, RAM improvements)" were valued by 3 users each, making them the most appreciated features. "Generative AI Wallpaper" was selected by 2 users, while another 2 users did not indicate any specific features as valuable. This visualisation shows a relatively even distribution in preferences, with some users valuing distinct features while others found no particular feature stood out.

Issues or Bugs with Features

Participants indicated whether they had encountered any issues or bugs with the custom ROM's features. Figure 11 shows that 80% of users had no issues or bugs during usage, while 20% reported issues.

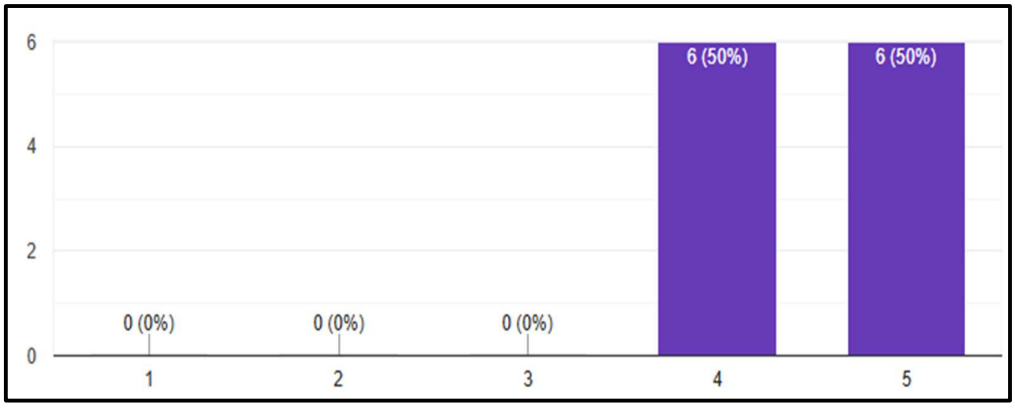


Figure 7: Performance rating of the custom ROM

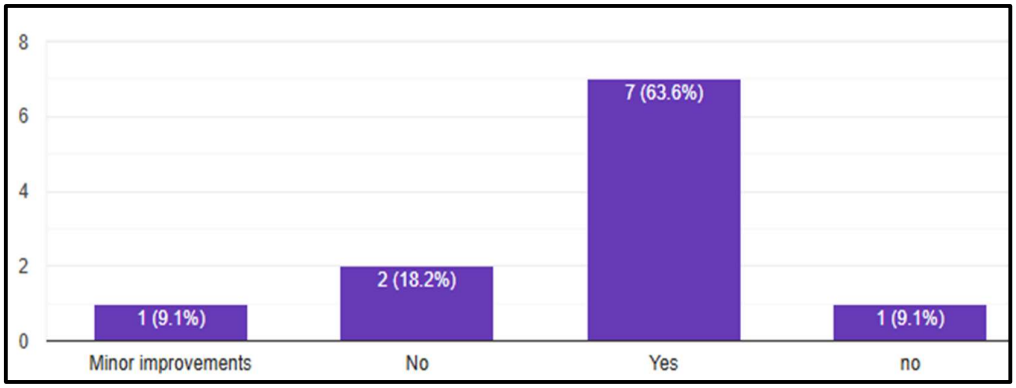


Figure 8: Improvements in battery life

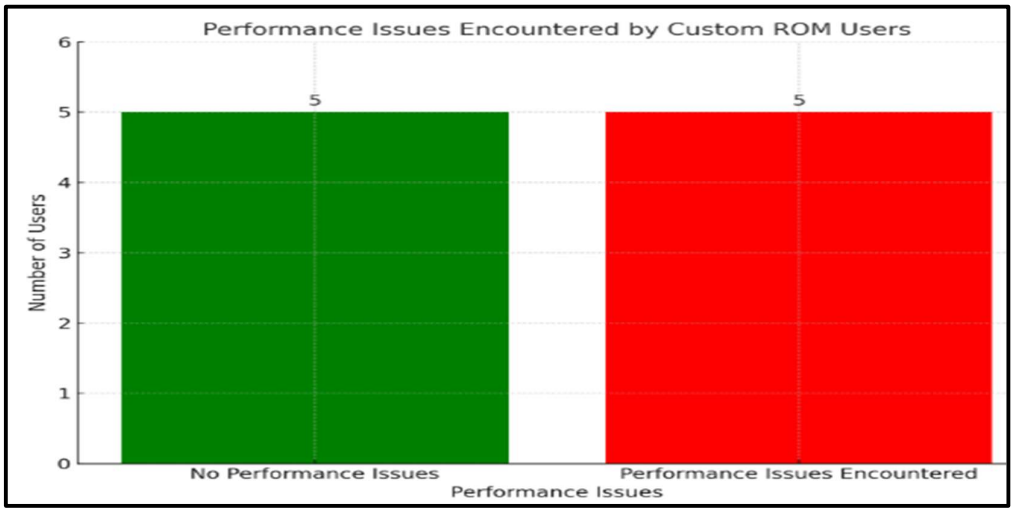


Figure 9: Performance issues

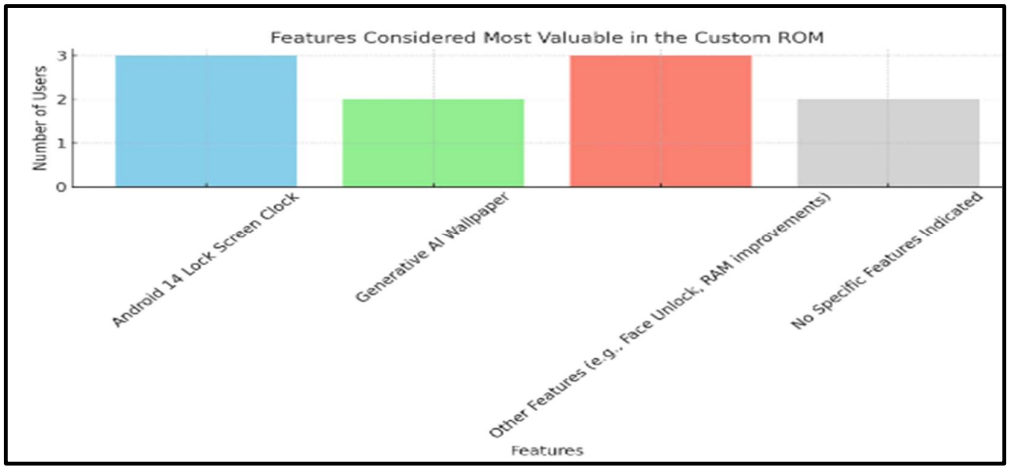


Figure 10: Most valuable features of the custom ROM

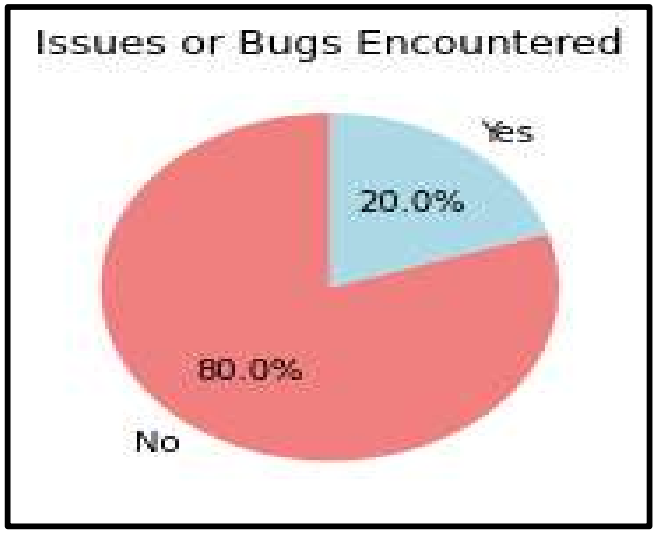


Figure 11: Issues or bugs encountered

User Experience Rating Compared to Stock ROM

Users rated the overall user experience compared to the stock ROM. Figure 12 shows that 40 % rated the custom ROM 5, while 60% rated 4, showing overall positive experiences.

Dislikes about the Custom ROM

Participants were asked to mention any aspects of the ROM's interface or user experience they did not like. Figure 13 shows that 4 users indicated no dislikes, 1 user indicated RAM issues, 2 users indicated too few features, 3 users indicated font and boot animation.

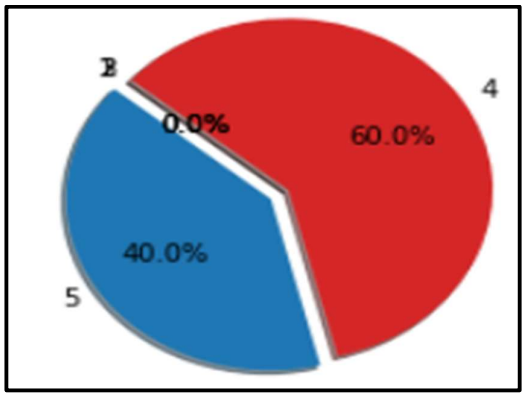


Figure 12: Experience rating for the custom ROM

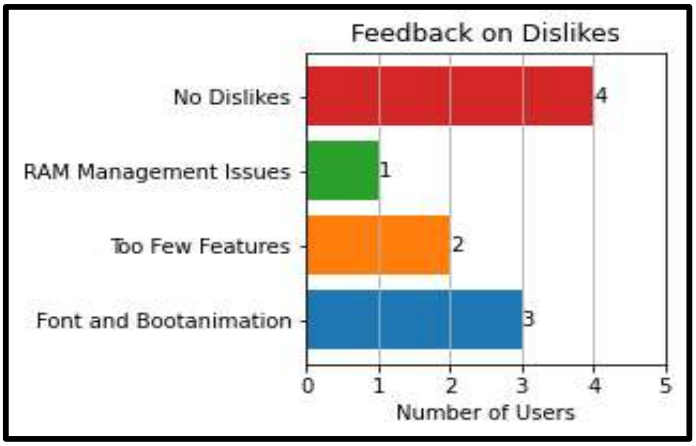


Figure 13: Feedback on dislikes

Likelihood to Recommend the Custom ROM

The likelihood to recommendation was rated on a Net Promoter Score (NPS) scale of 0-10. Figure 14 shows that 7 promoters had high satisfaction and indicated a strong likelihood to recommend the custom ROM. 3 users indicated moderate satisfaction with a potential to recommend while the result showed no users who saw the custom ROM as a detractor.

The user acceptance indicates that the majority of participants have over a year of experience with custom ROMs, with four users having more than five years of usage. Performance enhancements and access to the latest Android versions are the primary motivations for adopting custom ROMs. Users have generally observed improvements in battery life, with an average performance rating of 4.5 out of 5. Despite these benefits, half of the respondents reported occasional performance issues, such as crashes and slowdowns. Overall, the custom ROM received a high Net Promoter Score, with seven out of ten users likely to recommend it. By implementing this comprehensive testing and user acceptance strategy, we delivered a custom ROM that is not only feature-rich and performant but also stable, reliable, and user-friendly.

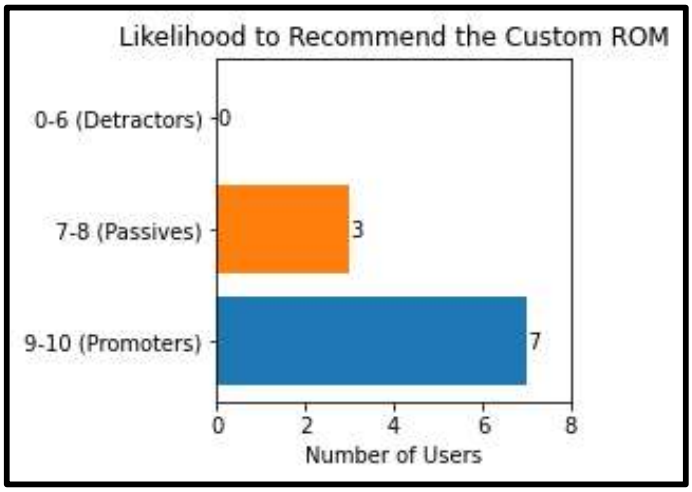


Figure 14: Recommendation possibilities

CONCLUSION

The study presented the development and evaluation of a custom Android ROM tailored for the Google Pixel 3, leveraging the Android Open-Source Project (AOSP) framework. Through targeted feature integration—including Android 14 lock screens, generative AI wallpapers, and enhanced system updates—the custom ROM demonstrated the potential to improve user experience by addressing the limitations of stock ROMs, such as lack of ongoing support and restricted customisation.

Performance evaluations using benchmark tools like AnTuTu and Geekbench indicated that the custom ROM offers competitive graphical capabilities, though CPU efficiency remains an area for further optimisation. While boot times and app launch speeds were slightly extended compared to stock Android 12, user acceptance testing showed favourable feedback for battery performance, usability, and design enhancements.

Although similar features can be accessed through newer Pixel models or official Android 15 builds, this work addresses a distinct challenge: enabling modern Android capabilities on unsupported legacy hardware. Hardware-level 5G integration is not feasible on the Pixel 3 due to the absence of a 5G modem, but the innovation lies in overcoming compatibility constraints, optimising system performance, and extending device viability through software-level engineering. By porting Android 14 to a device officially limited to Android 12, the project demonstrates how custom ROMs can offer practical alternatives in resource-constrained settings where upgrading hardware may not be feasible.

Future development can focus on optimising power management and resource allocation to refine overall performance. The study underscores the value of community-driven ROM development, highlighting custom ROMs as viable solutions for extending device usability, supporting digital sustainability, and enhancing the personalisation of legacy Android devices.

DECLARATIONS - ETHICS APPROVAL

This study involved an early-stage evaluation of the custom ROM with users to gather preliminary insights into its usability and performance. Due to time constraints, formal ethics approval was not obtained before conducting the evaluation. However, participation was entirely voluntary, and users were informed about the purpose of the study, their rights, and data confidentiality. No personally identifiable or sensitive data was collected. Future studies will undergo formal ethics review to ensure full compliance with institutional ethical guidelines.

ACKNOWLEDGEMENT

We want to express our deepest gratitude to Odunayo Abraham for his invaluable technical contributions to the development of the custom ROM. His expertise and dedication were instrumental in ensuring the project's success. Odunayo's innovative insights and unwavering support greatly enhanced the performance and functionality of the custom ROM. We are truly grateful for his commitment and hard work.

REFERENCES

Antutu. (2024). *Global top 10 performing IOS devices*. Antutu Benchmark.

Bissig, P., Brunner, G., Gubler, F., Wattenhofer, R., & Zingg, A. (2018). Towards measuring real-world performance of android devices. *2018 IEEE/ACS 15th International Conference on Computer Systems and Applications (AICCSA)*, 1–8.

Chahar, V., & Bhatia, P. K. (2025). Selection of effective parameters for optimizing software testing effort estimation. In *Applied Data Science and Smart Systems* (pp. 82–90). CRC Press.

Dharta, F. Y., Violin, V., Wahyono, T., & Putra, G. W. (2024). Analysis of User

Acceptance Levels Using the Unified Theory of Acceptance and Usage of Technology on Streaming Service and Video-On-Demand Channels Customers. *Jurnal Sistim Informasi Dan Teknologi*, 1–6.

El-Rewini, Z., & Aafer, Y. (2021). Dissecting residual APIs in custom android ROMs. *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*, 1598–1611.

Geekbench. (2024). *Introducing Geekbench6*. https://www.geekbench.com/

Khorikov, V. (2020). *Unit Testing Principles, Practices, and Patterns*. Simon and Schuster.

Meetkumar, R. P. (2022). Custom Android ROM's. *International Research Journal of Engineering and Technology (IRJET)*, *9*(3).

Mudit, M. J. (2023). Custom ROMs on Android Devices. *International Journal of Innovative Science and Research Technology*, *8*(7).

Pranjal, M., Saurav, M., Ankush, V., Sarang, U., & Nilima, J. (2019). Building Custom ROM Using AOSP and Improving Performance. *International Journal of Research and Analytical Reviews (IJRAR)*, *6*(1).

Rahimi, N., Nolen, J., & Gupta, B. (2019). Android security and its rooting—a possible improvement of its security architecture. *Journal of Information Security*, *10*(2), 91–102.

Santiago-Salazar, J. A., & Rico-Bautista, D. (2024). Clean Architecture: Impact on Performance and Maintainability of Native Android Projects. In M. Tabares, P. Vallejo, B. Suarez, M. Suarez, O. Ruiz, & J. Aguilar (Eds.), *Advances in Computing* (pp. 82–90). Springer Nature Switzerland.

Savola, R. M., Kylänpää, M., & Abie, H. (2020). Risk-driven security metrics for an Android smartphone application. *International Journal of Electronic Business*, *15*(4), 297–324.

Shreyas, S. (2020). Developing Custom ROM based on Android using AOSP. *International Journal for Research in Applied Science and Engineering Technology*, *8*, 709–714. https://doi.org/10.22214/ijraset.2020.30932

Suleman, M., Zhong, X., & Sun, Y. (2020). Empirical Research And Auxiliary Tool For Custom Android ROMs. *2020 International Symposium on Computer Engineering and Intelligent Communications (ISCEIC)*, 14–18. https://doi.org/10.1109/ISCEIC51027.2020.00011

Sutter, T. (2021). Firmwaredroid: Security analysis of the android firmware ecosystem. *ArXiv Preprint ArXiv:2112.08520*.

Taley, D. S., & Pathak, B. (2020). Comprehensive study of software testing techniques and strategies: a review. *Int. J. Eng. Res*, *9*(08), 817–822.